

ROADOCC LEARNS WHEN TO PERSIST, TRANSPORT, OR REFRESH MEMORY FOR ROADSIDE OCCUPANCY PREDICTION

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ABSTRACT

Fixed roadside cameras repeatedly observe a stable scene overlaid by sparse moving traffic. Temporal memory can recover weak observations, but reusing moving evidence at stale locations can corrupt occupancy predictions. Motion compensation addresses displacement, while reliance on the resulting history remains a separate learning problem. We introduce RoadOcc, which learns soft routing among fixed-coordinate history (*Persist*), velocity-addressed history (*Transport*), and current evidence (*Refresh*). Motion state and class-consistent historical support supervise these source choices. Dynamic-aware cross-attention (DCA) updates candidate locations, multi-scale voxel velocity estimation (VVE) constructs transport addresses from multi-scale current-history correspondence, and velocity-guided dynamic sparse fusion (VDSF) combines routed evidence under fixed sparse-token budgets. On InfraOcc, RoadOcc reaches 65.29 mIoU and 32.37 dynamic mIoU, gains of 4.44 and 4.71 over STCOcc. Controlled address experiments show that VVE raises dynamic mIoU by 0.87 over fixed-coordinate reading. Across three seeds, supervised P/T/R adds 1.40 dynamic points over motion-corrected retrieval, while removing Refresh costs 0.32 points. Results from two transfer models, Occ3D-nuScenes, and longer intervals provide additional support. Code will be released.

1 INTRODUCTION

Semantic occupancy prediction reconstructs free space, static layout, and traffic participants in a common voxel volume. Camera-only methods have made this representation practical through sparse voxel queries, BEV lifting, tri-plane features, and unified panoptic volumes (Cao & de Charette, 2022; Li et al., 2023a; Wang et al., 2023; Tian et al., 2023; Li et al., 2025b; Huang et al., 2023; Wei et al., 2023; Tang et al., 2024; Huang et al., 2024b). However, these methods are predominantly developed for moving platforms, where each observation is interpreted in an ego-centric frame. A roadside sensor instead repeatedly observes a persistent scaffold overlaid by sparse, short-lived dynamics. History is most valuable for weak or occluded actors, yet stale footprints can overwrite exactly these targets. The fixed world frame therefore shifts the central temporal problem to local evidence management. Static history is already aligned, whereas dynamic history must be selectively relocated or rejected.

The same history can provide either valid support or stale evidence. Fixed-coordinate memory preserves static layout and can recover weak or occluded vehicle boundaries, but it does not relocate an object that moved within the fixed coordinate system. Reading such memory from the former footprint can blur a dynamic boundary, duplicate an object, or attach it to nearby static layout. Discarding history avoids these errors but also removes useful support. The challenge is therefore not simply whether to use history, but to decide where temporal evidence is needed, where supporting history should be read, and how strongly to rely on each source. These cases can coexist within one scene or along a single object boundary, making a uniform temporal fusion rule insufficient.

Recent temporal methods demonstrate the value of multi-frame context for occupancy, flow, and 3D perception (Ma et al., 2024; Zheng et al., 2024; Wang et al., 2025; Chen et al., 2025a; Liao et al., 2025; Liu et al., 2025; Huang & Huang, 2022; Chang et al., 2024; Leng et al., 2025). Existing designs already incorporate temporal memory, motion alignment, and adaptive fusion. Motion alignment addresses displacement, but a flow-proposed address may still lack class-consistent support for the current voxel. A stationary object may reuse fixed-coordinate memory, a moving object may retrieve memory near $\mathbf{x} - \Delta t \mathbf{v}_t(\mathbf{x})$, and a newly appearing object may have to rely on current evidence. Adaptive weighting can represent these alternatives. Our focus is to supervise the source choice explicitly using motion state and semantic support, giving temporal fusion a structured learning target.

We therefore formulate temporal occupancy as explicit evidence-source routing. *Should each selected voxel Persist, Transport, or Refresh?* Persist reuses the fixed-coordinate source, Transport applies backwarp sampling to velocity-addressed history, and Refresh uses no history, only current evidence. Flow proposes where history should be read, while P/T/R learns preferences over the three sources. The supervision favors supported fixed history for stationary targets and supported transported history for moving targets, using Refresh when the corresponding candidate lacks support.

RoadOcc realizes this principle through a DCA-VVE-VDSF chain. DCA localizes the dynamic candidate and refines the current observation. VVE then combines current and temporal flow into dynamic flow, after which VDSF selects fixed-coordinate, backwarped, or current evidence. The stages therefore resolve where temporal reasoning is needed, where history should be read, and which source should enter fusion. Their intermediate outputs allow separate diagnostics of dynamic localization, historical addressing, and source selection. Figure 1 grounds the three evidence-source alternatives. Our contributions are summarized below.

- We propose RoadOcc, which formulates roadside temporal occupancy as a Persist–Transport–Refresh source-selection problem. Motion state and class-consistent support supervise the source preferences, complementing motion-based historical addressing.
- We implement this formulation in a sparse, multi-scale pipeline. DCA allocates image updates, VVE estimates transport addresses, and VDSF combines routed sources while retaining a short background memory path.
- We report joint occupancy and motion improvements on InfraOcc, three-seed routing controls, and module transfer to two temporal models. Occ3D-nuScenes and temporal-gap evaluations further test the design beyond the primary roadside setting and training cadence.

2 RELATED WORK

2.1 CAMERA OCCUPANCY AND ROADSIDE PERCEPTION

Camera occupancy methods lift multi-view image evidence into volumetric representations. MonoScene Cao & de Charette (2022) and VoxFormer Li et al. (2023a) establish image-based semantic scene completion, while BEVDet, BEVDepth, and BEVFormer improve multi-

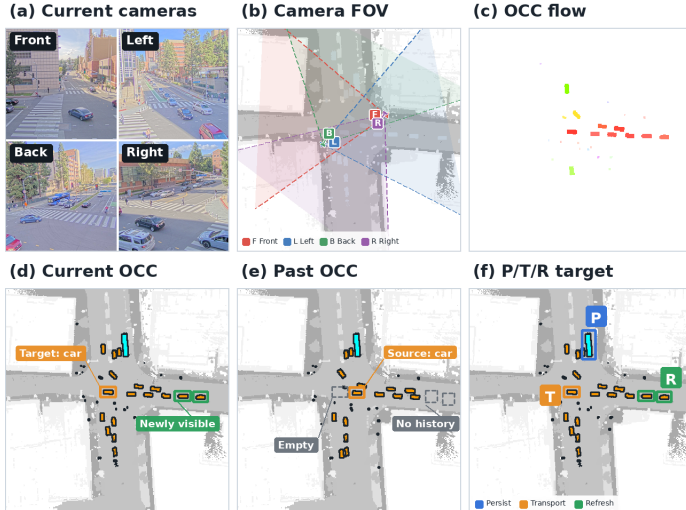


Figure 1: **Real-scene evidence for explicit source routing.** The top row shows current views, camera FOVs, and current-aligned occupancy flow (a–c). The bottom row shows current/past occupancy and P/T/R targets (d–f), illustrating a persistent bus, transported car, and newly visible cars assigned Refresh.

view lifting and BEV reasoning (Huang et al., 2021; Li et al., 2023b; 2025b). Occupancy-specific designs span tri-plane, dense, sparse, Gaussian, and forward-backward representations (Huang et al., 2023; Wei et al., 2023; Zhang et al., 2023; Tang et al., 2024; Huang et al., 2024b; Li et al., 2023c), as well as panoptic and instance-aware prediction (Wang et al., 2024; Li et al., 2025a). Unlike MC-BEVRO’s roadside BEV occupancy monitoring (Vaghela et al., 2025), RoadOcc routes evidence for class-aware 3D occupancy.

InfraOcc provides a roadside occupancy benchmark emphasizing persistent infrastructure (Yang et al., 2026). Figure 2 reveals its temporal asymmetry. Fixed roadway structure provides broad, stable support, while dynamic participants form sparse, short-lived trails. This asymmetry favors retaining static context while selectively relocating moving evidence.

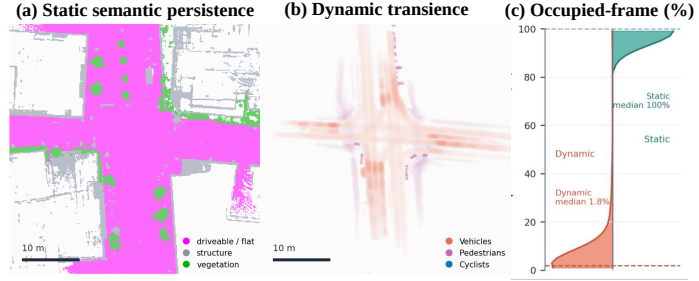


Figure 2: **Static-to-dynamic persistence in InfraOcc.**

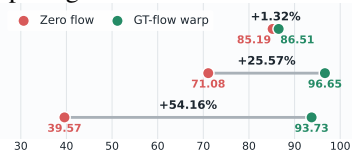
2.2 TEMPORAL OCCUPANCY, MOTION, AND ALIGNMENT

Temporal occupancy methods improve memory through recurrent aggregation, motion-aware alignment, and explicit association. Recurrent and video-based models maintain or forecast scene state (Huang & Huang, 2022; Chang et al., 2024; Huang et al., 2024a; Mohan et al., 2026), while occupancy-flow methods couple volumetric semantics with motion (Ma et al., 2024; Zheng et al., 2024; Wang et al., 2025; Chen et al., 2025a; Liao et al., 2025; Liu et al., 2025; Murhij & Yudin, 2024; Zheng et al., 2026). Other approaches organize temporal cues, learn semantic associations, or retrieve traversal priors (Leng et al., 2025; Chen et al., 2025b; Ouyang et al., 2025; Yuan et al., 2025). RoadOcc builds on temporal alignment and adaptive fusion by supervising a choice among fixed-coordinate history, transported history, and current evidence. Its emphasis is the motion- and support-based learning target for source selection, rather than the weighted-sum operation alone.

Table 1: **Generated GT-flow target at $t + 0.5$ s.** Zero-flow and GT-flow transport share the inter-frame transform and differ only by object displacement. Same/Dyn./Occ. denote unchanged semantics, dynamic class, and occupied class. Values are percentages and N counts valid queries in millions.

Speed	N (M)	Zero-flow transport			GT-flow warp			Δ Same
		Same	Dyn.	Occ.	Same	Dyn.	Occ.	
All	11.83	72.93	78.86	90.92	89.21	95.79	99.38	+16.28
Slow (< 2 m/s)	7.87	85.19	94.08	98.80	86.51	96.25	99.70	+1.32
Med. (2–5 m/s)	1.16	71.08	71.38	87.50	96.65	97.64	99.53	+25.57
Fast (≥ 5 m/s)	2.80	39.57	39.59	70.41	93.73	93.75	98.43	+54.16

Figure 3: **Motion compensation restores temporal consistency.** GT flow preserves dynamic support across speed ranges, whereas fixed-coordinate retrieval becomes increasingly misaligned as speed grows.



Flow-based alignment is closest to this problem. AsyncBEV aligns asynchronous cross-modal observations with BEV feature flow (Wang et al., 2026). In the fixed roadside setting, Table 1 and Figure 3 show that GT-flow warping improves same-class agreement by 1.32 points for slow objects, but by 25.57 and 54.16 points for medium and fast objects, motivating selective motion compensation. This is a diagnostic of the constructed targets. For candidates no faster than 2 m/s, low-speed target repair selects zero motion when its future same-class support score exceeds the transported score by at least 0.05.

3 METHOD

3.1 PRELIMINARY ANALYSIS

Let $\mathcal{I}_t = \{I_t^n\}_{n=1}^4$ denote the synchronized roadside images at time t . RoadOcc predicts semantic occupancy $\mathbf{y}_t$ over a voxel domain Ω and a planar velocity field $\mathbf{v}_t : \Omega \rightarrow \mathbb{R}^2$. Among the 18 occu-

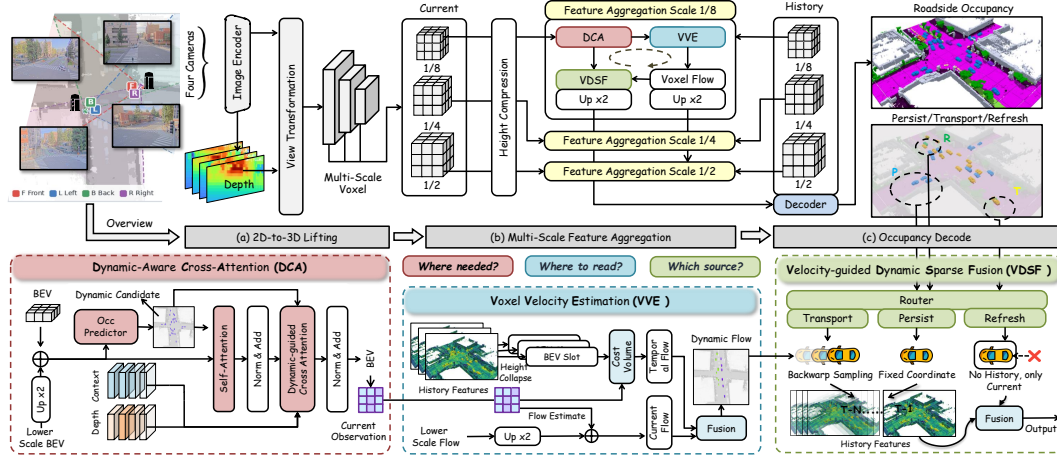


Figure 4: **RoadOcc framework.** The image encoder and view transformation construct multi-scale voxel features. At each feature aggregation scale, DCA produces the current observation, VVE fuses current, temporal, and lower-scale flow into dynamic flow, and VDSF routes fixed-coordinate, backwarped, or current evidence to fusion. Coarse-to-fine outputs feed occupancy decoding for semantic occupancy and voxel flow.

pancy states, the dynamic set is $\mathcal{C}_{\text{dyn}} = \{\text{bicycle, bus, car, motorcycle, pedestrian, truck}\}$. Velocity is expressed in the common roadside horizontal frame.

Because fixed roadside cameras share a persistent physical frame, cached history already lies on the current voxel grid at inference. We only reconcile relative BEV augmentations during training. Under a locally constant velocity approximation, history supporting a current voxel $\mathbf{x}$ is read from

$$\tilde{\mathbf{h}}_{t-\Delta t}(\mathbf{x}) = \text{Sample}(\bar{\mathbf{h}}_{t-\Delta t}, \mathbf{x} - \Delta t \mathbf{v}_t(\mathbf{x})), \quad (1)$$

where $\bar{\mathbf{h}}$ is same-grid history and Sample is trilinear interpolation. The three candidates are fixed-coordinate history at $\mathbf{x}$ (Persist), flow-addressed history at $\mathbf{x} - \Delta t \mathbf{v}_t(\mathbf{x})$ (Transport), and current evidence (Refresh). Equation 1 specifies an address, while VDSF learns how to combine these sources under motion- and support-based supervision.

3.2 OVERVIEW

RoadOcc decodes $\{\mathbf{V}_t^s\}_{s \in \{1/8, 1/4, 1/2\}}$ from coarse to fine. As shown in Figure 4, the encoder lifts four camera views into multi-scale voxel features. At each aggregation stage, DCA determines where evidence is needed, VVE determines where history should be read, and VDSF determines which source should enter fusion. The output feature and dynamic flow seed the next finer stage before occupancy decoding. Coarse predictions establish scene layout, while finer stages restore object boundaries and sparse traffic participants before the full-resolution output.

3.3 DYNAMIC-AWARE CROSS-ATTENTION (DCA)

Dynamic objects occupy few voxels, so uniformly repeating image cross-attention spends most computation on stable structure. After the first nonempty-guided image update, DCA combines the current representation with an upsampled lower-scale BEV to derive a dynamic candidate. Let $\pi_{t,\text{cur}}^s$ and $\pi_{t,\text{sta}}^s$ denote the provisional distributions produced by the current and static prediction heads after the first image update, and let $\mathcal{C}_{\text{sta}}^+$ contain the static classes and free space. The mean semantic discrepancy $\delta_{t,\text{sta}}^s$ over $\mathcal{C}_{\text{sta}}^+$ and current dynamic probability $p_{t,\text{dyn}}^s$ form the candidate map

$$c_t^s(\mathbf{x}) = \text{Norm}(\max\{\delta_{t,\text{sta}}^s(\mathbf{x}), p_{t,\text{dyn}}^s(\mathbf{x})\}). \quad (2)$$

Here Norm divides by the per-sample spatial maximum, clamped to 10^{-6} . Discrepancy recovers departures from the static hypothesis, while dynamic probability retains visible actors. Their maximum also preserves early semantic misses detected through change evidence.

The candidate map controls a second image cross-attention rather than directly reweighting the voxel feature. Following projected deformable attention (Li et al., 2025b) and occupancy-aware reference sampling (Liao et al., 2025), let $\mathbf{q}_{t,\mathbf{x}}^s$ be the BEV query at $\mathbf{x}$, $\mathcal{R}_n(\mathbf{x})$ its visible height anchors in camera n , and $\mathbf{F}_t^n$ the corresponding image feature. DCA retains $\mathcal{R}_{n,c}^s(\mathbf{x}) = \{\mathbf{r} \in \mathcal{R}_n(\mathbf{x}) : u_{\mathbf{r}} < c_t^s(\mathbf{r})\}$ and updates the query as

$$\mathbf{q}_{t,\mathbf{x}}^{s,+} = \mathbf{q}_{t,\mathbf{x}}^s + \mathbf{W}_o \frac{1}{|\mathcal{V}_{\mathbf{x}}|} \sum_{n \in \mathcal{V}_{\mathbf{x}}} \sum_{\mathbf{r} \in \mathcal{R}_{n,c}^s(\mathbf{x})} \beta_{n,\mathbf{r}}^s \mathcal{A}_{\text{def}}(\mathbf{q}_{t,\mathbf{x}}^s, \mathcal{P}_n(\mathbf{r}), \mathbf{F}_t^n), \quad (3)$$

where $\mathbf{r} = (\mathbf{x}, z_{\mathbf{r}})$ indexes a voxel-height score shared across cameras, $\mathcal{V}_{\mathbf{x}}$ contains valid cameras, and $\mathcal{P}_n$ projects anchors. Depth consistency $\beta_{n,\mathbf{r}}^s$ multiplies normalized deformable weights before camera averaging. During training, $u_{\mathbf{r}}$ is sampled from a normal distribution with mean 0.5 and standard deviation 1, truncated to $[0, 1]$. At inference, it is fixed to 0.5. The stochastic rule exposes uncertain boundary voxels during training, while deterministic thresholding keeps inference stable. Queries without retained support follow the residual path unchanged.

The candidate gates a second cross-attention over image depth context and also allocates VDSF tokens. DCA therefore localizes where additional image and temporal evidence is needed without repeatedly updating the stable scene.

3.4 MULTI-SCALE VOXEL VELOCITY ESTIMATION (VVE)

DCA refines the current observation but does not locate displaced history. VVE estimates its address from the DCA feature and nearest valid history. From the current occupancy distribution $\pi_t^s(\mathbf{x})$, it computes dynamic support $d_t^s = \sum_{c \in \mathcal{C}_{\text{dyn}}} \pi_{t,c}^s$ and nonempty support $e_t^s = 1 - \pi_{t,\text{free}}^s$. Here d_t^s guides height compression into the current BEV slot $\mathbf{B}_t^s$, while e_t^s guides temporal token selection. Applying the same compression to history gives $\bar{\mathbf{B}}_{t-\Delta t}^s$.

VVE next compares each current BEV slot with a small neighborhood in the historical slot, producing the local cost volume

$$\mathbf{K}_t^s(\mathbf{x}, \delta) = \left\langle \frac{\mathbf{B}_t^s(\mathbf{x})}{\|\mathbf{B}_t^s(\mathbf{x})\|_2}, \frac{\bar{\mathbf{B}}_{t-\Delta t}^s(\mathbf{x} + \delta)}{\|\bar{\mathbf{B}}_{t-\Delta t}^s(\mathbf{x} + \delta)\|_2} \right\rangle, \quad \delta \in [-r, r]^2, \quad (4)$$

where $r = 2$ yields a 5×5 neighborhood and both norms use an $\epsilon = 10^{-6}$ floor. The cost-volume decoder converts this local correspondence evidence into temporal flow.

VVE estimates flow recursively rather than solving each resolution in isolation. Let s^- denote the preceding coarser stage and set the upsampled lower-scale flow $\mathbf{v}_{t,\uparrow}^s$ to zero at $1/8$ and to $\text{Up}_s(\mathbf{v}_{t,\text{cur}}^{s-})$ otherwise. This prior initializes the current flow. Current appearance and its semantic embedding $\mathbf{S}_t^s$ predict a current-only correction, while the history feature and cost volume predict the temporal flow. Their fusion is

$$\begin{aligned} \mathbf{v}_{t,\text{cur}}^s &= \mathbf{v}_{t,\uparrow}^s + \Delta \mathbf{v}_{t,\text{cur}}^s, & \mathbf{v}_{t,\text{tmp}}^s &= f_{\text{tmp}}(\mathbf{B}_t^s, \bar{\mathbf{B}}_{t-\Delta t}^s, \mathbf{K}_t^s), \\ \hat{\mathbf{v}}_t^s &= (1 - \alpha_t^s) \mathbf{v}_{t,\text{cur}}^s + \alpha_t^s \mathbf{v}_{t,\text{tmp}}^s + \Delta \mathbf{v}_{t,\text{ref}}^s. \end{aligned} \quad (5)$$

Here f_{tmp} is the cost-volume decoder and $\alpha_t^s = 0.5 \mathbf{1}_{\text{hist}} \max_z d_t^s$ gates history, becoming zero at sequence starts. The residual $\Delta \mathbf{v}_{t,\text{ref}}^s$ produces the dynamic flow, which is lifted across height as the VDSF address. Coarse stages capture motion structure and fine stages refine boundaries.

3.5 VELOCITY-GUIDED DYNAMIC SPARSE FUSION (VDSF)

Velocity-guided sparse fusion. Applying the full temporal memory after DCA and VVE would mostly repeat static context. VDSF makes gather–fuse–scatter (Liao et al., 2025) motion aware by combining DCA guidance with nonempty support under a fixed token budget

$$\mathbf{g}_t^s = \text{clip}(c_t^s + \eta e_t^s, 0, 1), \quad \mathcal{X}_t^s = \text{TopK}(\mathbf{g}_t^s), \quad (6)$$

where η balances change evidence and occupied support. The selected set $\mathcal{X}_t^s$ receives full stage-local history. Its complement uses a short fixed-coordinate path, preserving background context at lower cost.

Table 2: Main InfraOcc occupancy results.

Method	gIoU	mIoU			Bike		Bus	Car		Motorcycle	Pedestrian	Truck	Other	Barrier	Cone	Driveable	Sidewalk	Terrain	Manmade	Vegetation	Free
		all	dyn	sta																	
BEVDet '21	68.62	45.14	21.19	63.11	9.23	42.75	38.04	9.68	15.01	12.45	42.88	70.36	53.77	72.04	68.07	73.09	64.28	60.38	94.87		
BEVFormer '22	81.72	47.76	10.73	75.53	7.97	20.39	17.65	3.47	8.62	6.30	42.47	76.26	72.83	85.19	81.95	88.95	79.52	77.05	97.56		
BEVDepth '23	71.89	46.81	19.64	67.19	14.84	33.87	38.74	3.42	16.86	10.14	48.73	71.05	62.37	73.31	69.32	76.07	70.51	66.15	95.54		
TPVFormer '23	73.80	43.09	4.36	72.13	0.10	3.69	13.80	0.41	6.86	1.33	42.86	75.29	69.69	80.61	77.62	86.63	72.61	71.70	96.12		
SurroundOcc '23	83.78	50.77	11.37	80.32	12.66	21.90	18.54	2.06	9.50	3.55	65.09	79.04	72.70	86.38	83.99	91.10	83.16	81.09	97.80		
CONet '23	78.56	49.25	18.93	72.00	3.06	37.24	39.47	4.42	17.64	11.72	45.01	75.31	64.94	84.29	79.45	84.27	72.96	69.79	97.08		
ALOCC '25	75.66	43.31	5.86	71.40	8.41	4.04	8.52	0.00	4.48	9.71	63.90	68.97	63.40	90.09	77.77	78.16	65.07	63.85	97.12		
Let Occ Flow '24	87.18	51.38	8.78	83.34	11.27	11.53	14.07	4.34	8.12	3.34	62.06	75.06	85.56	92.55	85.63	92.98	87.10	85.74	98.56		
SparseOcc '24	71.61	45.42	9.41	72.42	1.50	20.94	24.87	0.00	4.60	4.57	67.68	76.37	61.51	74.67	76.59	88.54	68.79	65.19	96.31		
GaussianFormer '24	77.89	43.63	10.45	68.51	3.13	22.11	23.17	1.29	8.21	4.80	36.70	68.54	55.60	82.59	79.37	74.20	77.85	73.21	96.71		
CRT-Fusion-S '24	88.73	55.94	21.98	81.40	13.02	45.16	38.48	7.34	18.99	8.90	63.53	71.99	74.82	93.38	84.40	91.19	86.39	85.53	98.73		
STCOcc-S '25	78.54	50.71	23.48	71.14	13.43	47.83	39.62	10.21	17.83	11.93	50.18	71.54	58.33	90.30	78.78	84.90	67.72	67.32	97.47		
CRT-Fusion '24	80.35	53.83	27.02	73.94	32.58	46.63	39.49	12.22	20.81	10.41	53.08	72.27	68.15	89.06	79.09	84.48	72.06	73.30	97.68		
STCOcc '25	92.54	60.85	27.66	85.74	27.13	48.78	44.93	10.23	22.85	12.03	68.84	70.93	81.60	95.88	89.41	95.36	91.85	92.04	99.18		
RoadOcc (ours)	94.03	65.29	32.37	89.98	37.60	55.46	49.36	13.01	22.66	16.13	87.66	72.05	86.03	97.85	92.81	96.90	92.91	93.66	99.35		

Each scale receives multi-frame history features and their VVE motion priors. For $\mathbf{x} \in \mathcal{X}_t^s$, elapsed time converts dynamic flow into a voxel offset. Backwarp sampling produces the Transport candidate $\mathbf{h}^T$, while the fixed-coordinate read produces the Persist candidate $\mathbf{h}^P$.

Persist-Transport-Refresh routing. Dynamic flow proposes where to read history, but a corrected address may still lack semantic support. VDSF therefore decides which source should enter fusion. It retains $\mathbf{h}^P$ and $\mathbf{h}^T$ and defines the current-only Refresh candidate as $\mathbf{h}^R = \mathbf{q}_t$. During training, the route target is constructed from the current and nearest historical semantic occupancy labels together with the motion target, and assigns each current dynamic voxel according to its motion state and class-consistent historical support as follows.

$$z_t(\mathbf{x}) = \begin{cases} P, & \|\mathbf{v}_t(\mathbf{x})\| \leq \epsilon \wedge \text{Supp}(\mathbf{x}), \\ T, & \|\mathbf{v}_t(\mathbf{x})\| > \epsilon \wedge \text{Supp}(\mathbf{x} - \Delta t \mathbf{v}_t(\mathbf{x})), \\ R, & \text{otherwise,} \end{cases} \quad (7)$$

where Supp tests same-class occupancy at the same height within the local correspondence radius. We use $\epsilon = 10^{-3}$ m/s and a one-voxel radius to absorb discretization error. The construction follows the semantic occupancy label space used by InfraOcc and is evaluated at every execution scale using its physical cell size, with a native-grid endpoint for full-resolution supervision. The motion target selects which historical candidate is tested according to motion state. The fixed-coordinate candidate is used for stationary voxels and the transported candidate for moving voxels. If that candidate lacks support, the target is Refresh, even if the alternative historical address has same-class support. This is a motion-conditioned source preference. Targets use GT motion, while inference uses predicted VVE motion and learned route probabilities. Future occupancy is used only for offline target construction, never as model input.

At each feature aggregation scale, the router combines the current observation, nearest fixed-coordinate history feature, normalized dynamic flow, and upsampled coarser P/T/R distribution. It predicts stage-local logits. A lightweight native head upsamples the finest context with full-resolution flow. Softmax gives $\alpha_t^s = (\alpha_P, \alpha_T, \alpha_R)$ and the routed candidate

$$\mathbf{h}_t^{\text{PTR}} = \alpha_P \mathbf{h}^P + \alpha_T \mathbf{h}^T + \alpha_R \mathbf{h}^R, \quad (\alpha_P, \alpha_T, \alpha_R) = \text{softmax}(\psi(\cdot)). \quad (8)$$

P/T/R therefore compete as one route distribution rather than independent weights. Transport uses backwarp sampling, Persist uses the fixed-coordinate source, and Refresh uses no history, only current evidence. Fusion uses soft probabilities and reserves argmax for diagnostics.

Let $\mathbf{H}_t^{\text{PTR},s}(\mathbf{x})$ concatenate the routed multi-frame history at scale s . VDSF combines this temporal evidence with the current observation and its semantic embedding through the fusion MLP

$$\mathbf{q}_{t,\text{fuse}}^s(\mathbf{x}) = \mathbf{q}_t^s(\mathbf{x}) + \Phi_s([\mathbf{q}_t^s(\mathbf{x}), \mathbf{H}_t^{\text{PTR},s}(\mathbf{x}), \text{Emb}(\pi_t^s(\mathbf{x}))]). \quad (9)$$

The output residual is scattered to $\mathcal{X}_t^s$ and joined with the background path. The fused volume and dynamic flow feed the next scale and are retained for subsequent frames. The three stages use $K = [2000, 512, 128]$ sparse tokens and retain $[8, 4, 2]$ history samples, respectively; a historical source is valid when it has same-class support at the same height within a 3×3 XY neighborhood.

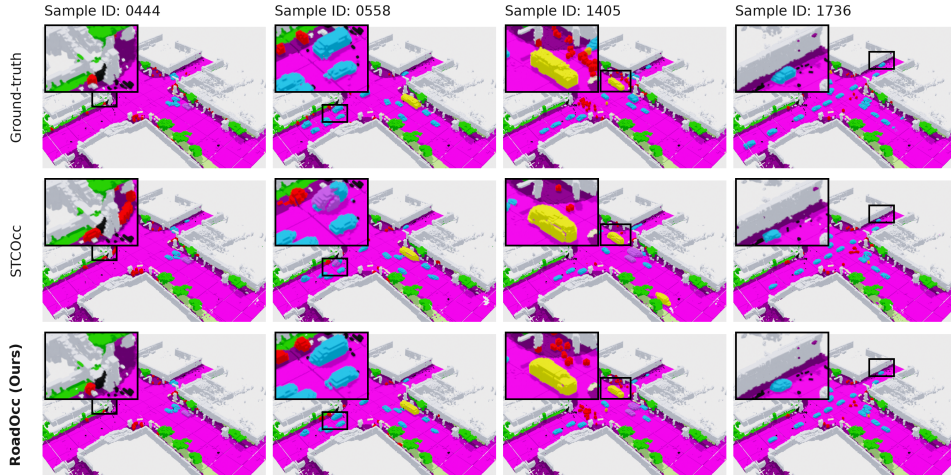


Figure 5: **Qualitative comparison with STCOcc.** RoadOcc more closely recovers dynamic-object presence, semantic identity, and geometry, including spurious pedestrian occupancy, vehicle-class confusion, bus-shape distortion, and an incomplete car footprint.

Table 3: **Joint occupancy–velocity comparison.**

Method	Occupancy					Velocity		Diagnostics	
	mIoU	Dyn.	Sta.	gIoU	RayIoU	Direct	MAVE	TP-MAVE	DSR
CRT-Fusion (NIPS 2024)	53.83	27.02	73.94	80.35	48.08	2.227		1.839	57.56
+ VDSF P/T/R	57.15	28.30	78.79	83.46	52.34	1.990		1.958	59.45
Δ	+3.32	+1.28	+4.85	+3.11	+4.26	−0.237		+0.119	+1.89
STCOcc (CVPR 2025)	60.85	27.66	85.74	92.54	55.94	1.946		0.703	57.12
+ VDSF P/T/R	61.81	28.80	86.56	93.59	58.07	1.770		0.648	58.78
Δ	+0.96	+1.14	+0.82	+1.05	+2.13	−0.176		−0.055	+1.66
RoadOcc (ours)	65.29	32.37	89.98	94.03	65.44	1.669		<u>0.661</u>	61.11

3.6 TRAINING OBJECTIVE

Training jointly optimizes four functional objectives.

$$\mathcal{L} = \lambda_{\text{depth}} \mathcal{L}_{\text{depth}} + \mathcal{L}_{\text{sem}} + \lambda_{\text{mot}} \mathcal{L}_{\text{mot}} + \lambda_{\text{route}} \mathcal{L}_{\text{route}}. \quad (10)$$

$\mathcal{L}_{\text{depth}}$ supervises camera depth for view transformation, while $\mathcal{L}_{\text{sem}}$ covers full- and multi-scale occupancy together with the static, dynamic, and DCA auxiliaries. $\mathcal{L}_{\text{mot}}$ supervises the dynamic flow used for backwarp sampling, whereas $\mathcal{L}_{\text{route}}$ supervises the P/T/R router on valid dynamic voxels. Together, they separate where history features are read from which source enters fusion. We set $\lambda_{\text{depth}} = 0.5$, $\lambda_{\text{mot}} = 0.1$, and $\lambda_{\text{route}} = 0.5$. Multi-scale semantic and route losses use relative weights $[1, 0.5, 0.25, 0.125]$ from native to $1/8$ resolution.

4 EXPERIMENTS

Dataset and metrics. We evaluate camera-only roadside occupancy on InfraOcc (Xiang et al., 2024; Yang et al., 2026). Four synchronized views are mapped to 18 occupancy states. Under the InfraOcc protocol, semantic mIoU excludes construction vehicle, trailer, other flat, and Free. Free IoU is reported separately in the class-wise table. We report semantic and geometric IoU, next-frame dynamic IoU, and motion coverage. Direct MAVE covers all GT dynamic voxels, while TP-MAVE conditions on semantic matches and is paired with recall (DSR). InfraOcc is the primary fixed-roadside protocol. We additionally evaluate module transfer, Occ3D-nuScenes, and temporal-gap changes. The benchmark contains 290 temporally continuous roadside sequences: 215 for training and 75 for evaluation.

Implementation details. RoadOcc uses a ResNet-50 backbone, 256×704 images, and three aggregation scales at 3.2, 1.6, and 0.8 m, followed by 0.4 m output voxels over $[-64, 64] \times [-64, 64] \times [-4.8, 1.6]$ m. Frames are 0.5 s apart. AdamW training runs for 24 epochs from 5×10^{-4} with

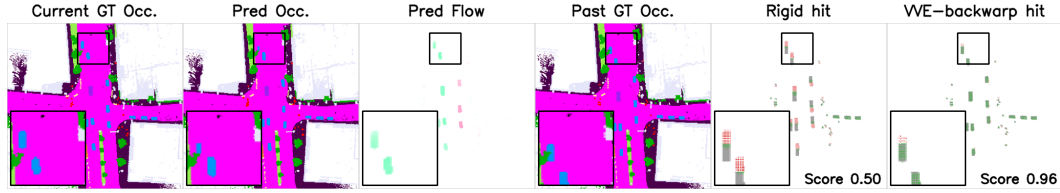


Figure 6: **Velocity-guided historical retrieval.** With queried dynamic targets fixed by current GT, the panels show the end-to-end occupancy prediction and compare a fixed-coordinate read with the VVE backtrace into aligned past GT. Black boxes are enlarged in the insets. ■ same-class matches, ■ unmatched targets, and ■ other queried voxels are shown in green, red, and gray, respectively. The score is the same-class hit rate within the one-voxel matching radius.

10^{-2} weight decay and cosine annealing. Comparisons share the voxel range, ignored classes, and evaluator. Controlled variants keep the backbone, data split, optimizer and other configuration fixed.

4.1 MAIN RESULTS

InfraOcc occupancy benchmark. Table 2 tests whether the proposed routing chain improves occupancy under the InfraOcc evaluator. RoadOcc achieves the highest mIoU among the compared methods at 65.29, a +4.44 gain over STCOcc. Dynamic mIoU rises by +4.71 to 32.37, while static mIoU rises by +4.24 to 89.98. Five of six dynamic classes improve, led by Bike (+10.47) and Bus (+6.68), whereas Pedestrian changes by -0.19 . Thus the aggregate gain covers both traffic participants and persistent structure, with variation by class.

Joint occupancy-velocity behavior. Table 3 evaluates motion accuracy and semantic coverage alongside occupancy. Direct MAVE measures velocity error over all GT dynamic voxels, while DSR measures dynamic semantic recall. Relative to STCOcc, RoadOcc lowers Direct MAVE from 1.946 to 1.669 and raises DSR from 57.12 to 61.11, improving motion accuracy alongside dynamic recovery. Adding VDSF P/T/R to CRT-Fusion and STCOcc likewise raises Dyn. by +1.28 and +1.14, respectively, while improving Direct MAVE and DSR. The consistent trend supports the module’s utility beyond the full RoadOcc architecture. TP-MAVE conditions on semantic matches and must be read with DSR. For CRT-Fusion, the +1.89 DSR gain indicates broader dynamic recovery even though the conditional error over matched voxels increases.

Qualitative occupancy recovery. Figure 5 supports the aggregate dynamic gain with representative matched scenes. STCOcc produces stale or spurious pedestrian occupancy, confuses a vehicle class, distorts a bus footprint, and incompletely recovers a car. RoadOcc more closely recovers the GT class and extent in these examples while preserving the roadway.

Transfer and temporal gaps. On mask-trained Occ3D-nuScenes, RoadOcc reaches 45.01 overall and 39.80 dynamic mIoU, improving over STCOcc by 0.41 and 0.79 points, respectively. Without fine-tuning the 0.5 s-trained checkpoints, testing at 1.5 s yields 33.49 Dyn. for RoadOcc and 28.58 for STCOcc on the same 1,520 anchors. These results extend the empirical support beyond the primary dataset and training cadence. They evaluate the complete model rather than isolate P/T/R.

Table 4: **Component ablation.**

Base	Design			Output ↑		
	DCA	VVE	VDSFP/T/R	All	Dyn.	Nxt.
✓				60.85	27.66	26.74
✓			✓	61.05	28.01	26.99
✓	✓			61.69	28.55	27.64
✓	✓		✓	63.01	30.10	29.44
✓	✓	✓	✓	64.17	30.97	30.78
✓	✓	✓	✓	65.29	32.37	31.72

4.2 ABLATION STUDIES

Using Liao et al. (2025) as the baseline, we hold the backbone, token budget, temporal-memory configuration, and training protocol fixed unless explicitly varied. Tables 4, 5, and 6 then examine component composition, historical-address correction, and P/T/R source selection, respectively.

Overall ablation. Table 4 follows the routing chain. VDSF alone adds only 0.20 mIoU. Adding VDSF to the DCA variant raises Dyn. by +1.55 and Nxt. by +1.80. VVE then adds +1.16 mIoU,

and P/T/R adds a further +1.12 mIoU. This progression supports combining candidate selection, motion-based retrieval, and source routing. Tables below examine latter two design in more detail.

Address correction and source selection. With capacity and token budget fixed, the fixed-coordinate and VVE reads in Table 5 differ only in historical address. VVE raises Dyn. by +0.87, lowers dMAVE from 2.146 to 1.690, and raises DSR by +1.51. P/T/R then adds +1.40 Dyn. and +1.76 DSR, showing that learned source selection improves on motion-corrected retrieval.

P/T/R routing controls. Table 6 compares routing variants within DCA–VVE–VDSF. None omits P/T/R. Gate adaptively mixes the same candidates. State adds P/T/R supervision without executing its probabilities. Full uses the supervised routes in fusion. Full w/o R masks α_R and renormalizes (α_P, α_T). The complete control contract is given in the appendix. Across three seeds, Full reaches 32.37 ± 0.11 Dyn., compared with 30.97 ± 0.27 for None and 31.25 ± 0.20 for Gate. State and Full improve successively by 0.63 and 0.49 points over the preceding variant. The clearest gain is in dynamic occupancy. The dMAVE change from None to Full is smaller (0.021 m/s), with standard deviations of 0.019 and 0.020 m/s. Removing Refresh from Full reduces Dyn., Nxt., and DSR by 0.32, 0.31, and 0.42 points, respectively, while increasing dMAVE by 0.010 m/s. Under the same three-seed protocol, the 0.32-point Dyn. gap exceeds either reported run-to-run standard deviation (0.11 and 0.12). We treat the other metric changes as supporting trends rather than standalone significance claims.

Table 5: **Controlled address and source selection.** All variants use the same sparse-token budget and model capacity.

Variant	Design		Output			
	VVE	P/T/R	Dyn.↑	Nxt.↑	dM↓	DSR↑
Fixed read			30.10	29.44	2.146	57.84
VVE read	✓		30.97	30.78	1.690	59.35
Full	✓	✓	32.37	31.72	1.669	61.11

Table 6: **P/T/R routing controls.** Mean $\pm$ standard deviation is reported for the multi-seed variants. w/o R denotes full w/o Refresh.

Variant	Dyn.↑	Nxt.↑	DSR↑	dM↓
None	30.97 $\pm$.27	30.78 $\pm$.34	59.35 $\pm$.39	1.690 $\pm$.019
Gate	31.25 $\pm$.20	30.94 $\pm$.30	59.75 $\pm$.45	1.685 $\pm$.018
State	31.88 $\pm$.18	31.38 $\pm$.25	60.52 $\pm$.40	1.675 $\pm$.017
w/o R	32.05 $\pm$.12	31.41 $\pm$.18	60.69 $\pm$.05	1.679 $\pm$.033
Full	32.37$\pm$.11	31.72$\pm$.26	61.11$\pm$.38	1.669$\pm$.020

Routing scope. Equation 7 supervises motion-conditioned semantic preferences, not instance correspondence. Labels use the nearest history, and each scale shares its predicted route across valid memory slots. GT-conditioned routing visualizations separate source choices from occupancy detection. They do not measure candidate misses or guarantee support at every predicted historical address. Dyn., Nxt., Direct MAVE, and DSR provide complementary end-to-end metrics.

Visual analysis. Figure 6 compares zero-motion reading with VVE backtracing. Current GT fixes the queried dynamic voxels and their classes, while predicted occupancy provides the end-to-end context. Red-to-green changes show dynamic flow relocating queries through backwarp sampling onto same-class past GT, isolating address quality from the subsequent routing decision.

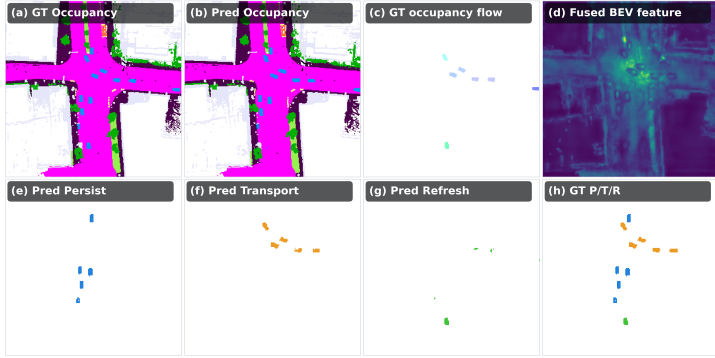


Figure 7: **Predicted P/T/R on GT dynamic support.**

On the same GT dynamic support, Figure 7 then shows Persist following fixed-coordinate support, Transport following displacement, and Refresh favoring current evidence. Together, the visualizations inspect historical addresses and learned source preferences on dynamic targets.

5 CONCLUSION

In this work, we present RoadOcc, a voxel-level evidence-source routing framework for temporal roadside occupancy. Beyond implicit temporal aggregation, RoadOcc explicitly learns where history should be retrieved and how strongly it should be reused. The proposed DCA uses current image evidence to identify motion-sensitive voxels, while VVE estimates their historical source addresses,

and VDSF combines evidence through Persist, Transport, or Refresh routing. On the InfraOcc protocol, the unified formulation achieves the highest occupancy accuracy among the compared methods while improving dynamic recovery and velocity estimation. Extensive experiments further demonstrate the superiority of our design. Future work will extend this principle to broader moving-platform settings, longer temporal horizons, and instance-level modeling.

AI USE STATEMENT

Generative AI tools were used for literature organization, reviewer-style feedback, and language editing. The authors verified all citations, technical descriptions, analyses, and revisions, and take responsibility for the final content of this work.

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